DQH-BAN: Dual-Stream Quantum Healthcare Intelligence Architecture (DQH-BAN-EXP-2026-V2)

Abstract: We present DQH-BAN, a dual-stream quantum healthcare intelligence architecture integrating quantum-assisted cybersecurity (Stream A) and physiological signal intelligence (Stream B) anchored by a Hyperledger Fabric security orchestration layer. All metrics reported in this paper are generated directly from verified offline checkpoints across seven physical benchmarks.

Dataset	Model	Acc (%)	Recall (%)	Macro-F1 (%)	FPR (%)	FNR (%)
CICIoMT2024	XGBoost	96.24	96.06	85.71	0.57	3.94
WUSTL-EHMS-2020	XGBoost	97.43	87.62	94.03	1.17	12.38
WUSTL-HDRL-2024	XGBoost	99.81	99.55	99.39	0.17	0.45
CIC-IoT-2023	XGBoost	99.01	98.65	99.00	0.52	1.35
ToN-IoT	XGBoost	99.47	99.27	99.47	0.34	0.73
CICIoMT2024	classical_matched	96.24	96.75	84.38	13.00	3.25
CICIoMT2024	Quantum_4q_l2_ring	90.69	90.23	73.75	0.96	9.77
WUSTL-EHMS-2020	Quantum_4q_l2_ring	93.38	52.44	81.43	0.75	47.56
WUSTL-HDRL-2024	Quantum_4q_l2_ring	99.45	98.44	98.26	0.46	1.56
CIC-IoT-2023	Quantum_4q_l2_ring	98.34	98.34	85.94	1.69	1.66
ToN-IoT	Quantum_4q_l2_ring	99.31	58.57	84.57	0.14	41.43

Verification Certification: Passed Round 1 (16 Technical Checks) and Round 2 (End-to-End Clean Session Recalculation).